**PHYS1401 Final Exam Version 2 (105/100 pts)**

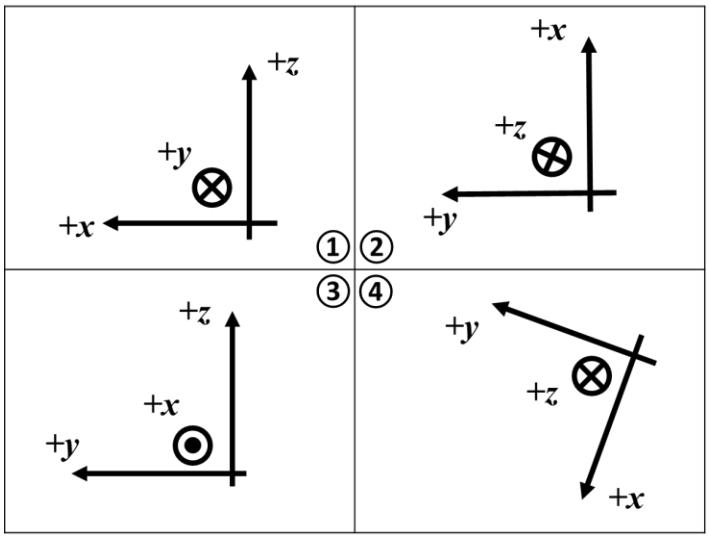
**NOTE(IMPORTANT!!): Please detach, sign, and submit the equation sheet to receive credit.**

**Print Name:**_____

**Student ID:**_____ **Course Section:**_____

**Part I: Multiple-choice Questions (Only the answer shall be credited. Each question has only one correct answer.)**

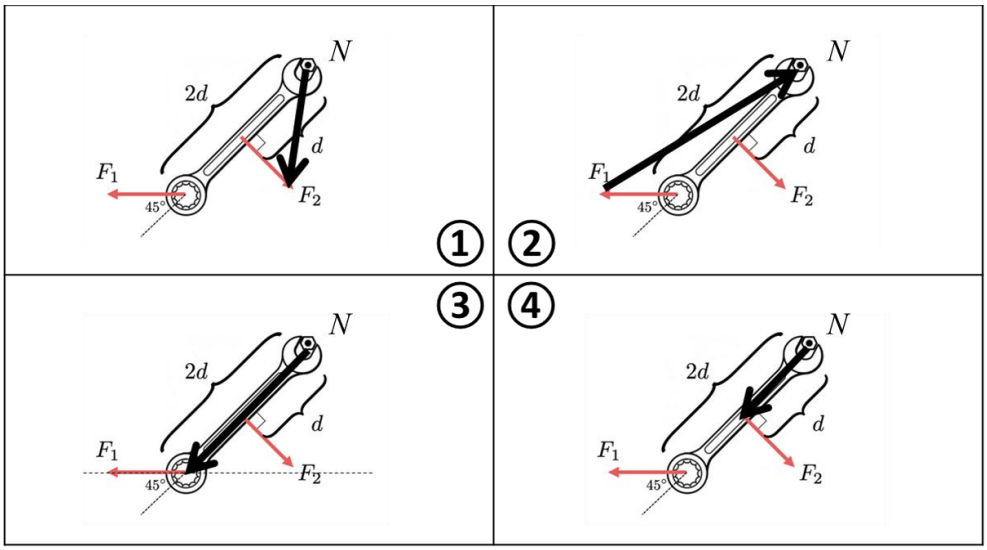
1. (5pts) Which of the diagrams below shows a correct right-hand 3D coordinate system (i.e., $\hat{x} \times \hat{y} = \hat{z}$)?




A. ①
B. ②
C. ③
D. ④

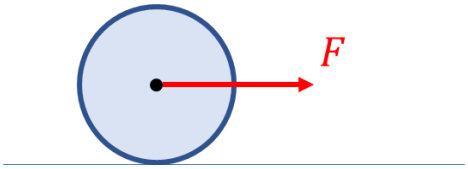
2. (5pts) A girl is at the carnival and rides the Merry-Go-Round. The ride moves with a constant angular velocity $\omega = 1.5$ rad/s. As the ride comes to an end, a constant angular acceleration of $\alpha = -0.3$ rad/s$^2$ is applied. How long does it take for the Merry-Go-Round to come to a stop?
   A. 3 s
   B. 4 s
   C. 5 s
   D. 6 s

3. (5pts) Two forces, $F_1$ and $F_2$ are applied to a wrench to exert torque on a nut (point $N$) at different distances and angles. Considering the torque equation $\vec{\tau} = \vec{r} \times \vec{F}$ for evaluating the torque on the nut due to $\vec{F}_2$, which of the following diagrams shows the correct position vector $\vec{r}$?



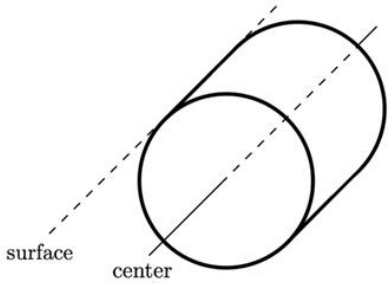

A. ①
B. ②
C. ③
D. ④

4. (5pts) A uniform disk is pulled by a force $F$ through a string attached to its center of mass. The disk rolls smoothly without slipping on a horizontal tabletop. Which of the following statements is correct regarding its rotational motion?



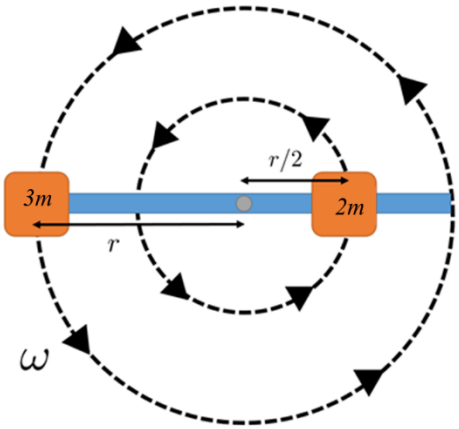

A. The torque applied to the center of the disk due to the normal force is zero.
B. The frictional force is negligible.
C. The torque around the center of the disk due to the force $F$ is not zero.
D. The torque around the center of the disk due to the normal force is not zero.

5. (5pts) A solid cylinder has a mass $M$ and radius $R$. The moment of inertia about its symmetrical axis is $\frac{1}{2}MR^2$. What is the moment of inertia of the cylinder about a parallel axis that lies on the surface of the cylinder?



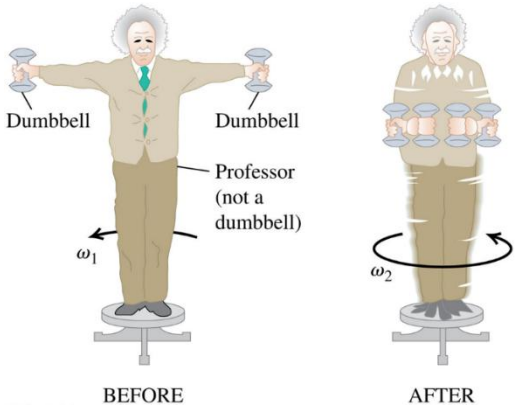

A. $\frac{5}{2}MR^2$
B. $\frac{3}{2}MR^2$
C. $MR^2$
D. $\frac{1}{2}MR^2$

6. (5pts) Two small square masses are connected to a massless rod. The whole configuration is rotating with a constant angular velocity $\omega$. What is the kinetic energy of the system?



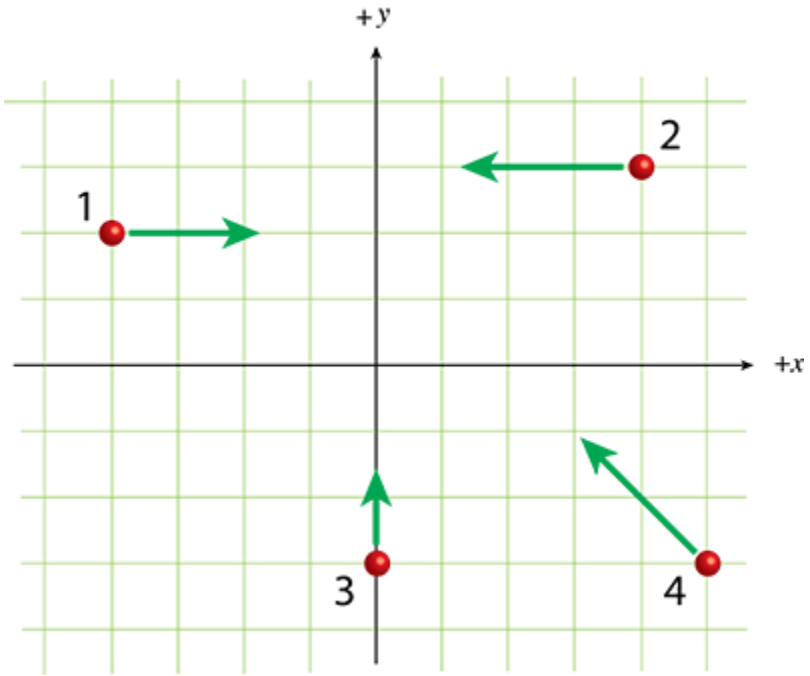

A. $\frac{9}{8}mr^2\omega^2$
B. $5mr^2\omega^2$
C. $\frac{4}{3}mr^2\omega^2$
D. $\frac{7}{4}mr^2\omega^2$

7. (5pts) A physics professor is standing on a frictionless rotating stool. The angular velocity is changed from $\omega_1$ to $\omega_2$ as he moves his arms closer to his chest. Which of the following statements is correct regarding the professor + stool system?




A. $\omega_1 = \omega_2$
B. The moment of inertia does not change.
C. The angular momentum does not change.
D. The angular momentum increases.

8. The diagram shows four particles (labeled 1, 2, 3, and 4) moving in the x-y plane. Particle 1 is located at (-4 m, 2 m) and is traveling in the positive x-direction. Particle 2 is positioned at (4 m, 3 m) and is moving in the negative x-direction. Particle 3 is at (0 m, -3 m) and is moving in the positive y-direction. Particle 4 is located at (5 m, -3 m) and has a velocity that makes an angle of 45 degrees with the coordinate axes. Which particle(s) have an angular momentum about the origin with a direction pointing into the page?



A. Particle 1.
B. Particles 2 and 4.
C. Particle 3.
D. None of the above.

**Questions 9-10 are based on the following scenario:** Two blocks on a horizontal, frictionless track are moving toward each other, as shown. Assume the collision between the blocks is perfectly elastic.




9. (5pts) What is the velocity of the center of mass of the two-block system?
   A. 0
   B. *v* to the left
   C. *v* to the right
   D. *v/4* to the left

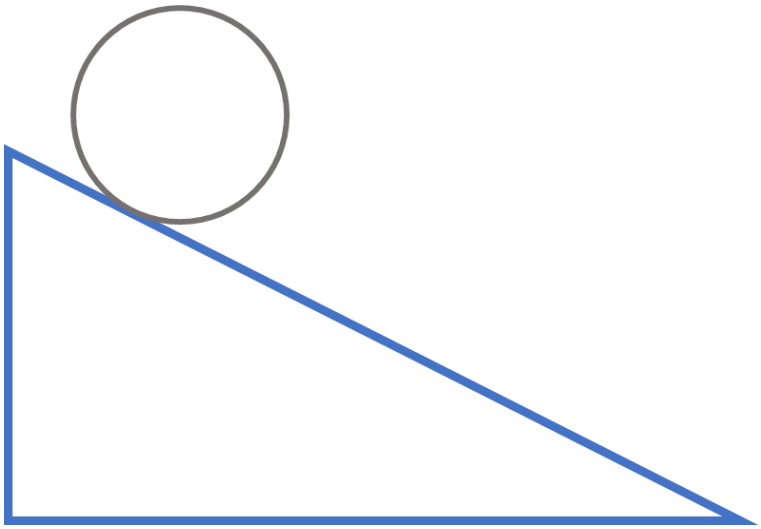
10. (5pts) In the center of mass reference frame, what is the velocity of the right block after the collision?
   A. 9*v*/4 to the left
   B. 9*v*/4 to the right
   C. 3*v*/4 to the left
   D. 3*v*/4 to the right

**Part II: Problem-solving Question (Please present the complete problem-solving procedure. Answers with no supportive calculation shall not be credited)**

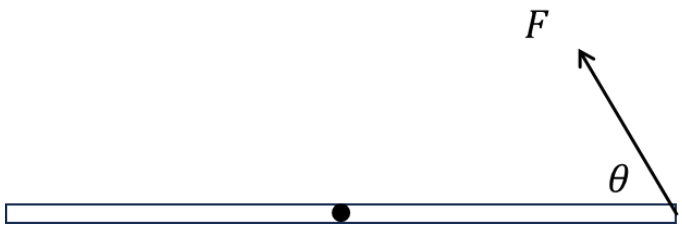
11. (15pts) A 510 g disk with a radius of 12 cm is supported by a pivot through its center. It is initially at rest then experiences a constant angular acceleration for 15 seconds, at the end of which it is spinning with a frequency of 85 revolutions per minute.
    a. (10pts) Calculate the magnitude of the angular velocity of the disk in radians per second at time $t$ = 15 seconds.

    b. (5pts) Calculate the magnitude of the net torque on the disk.

12. (20pts) A uniform disk with a radius of 4.3 cm and a mass of 270 g is released from rest on a ramp inclined at an angle 25° above the horizontal. The disk rolls without slipping, as shown in the diagram. Using the five-step problem-solving procedure, determine both the direction and magnitude of the frictional force exerted on the disk. As part of the solution, include a free-body diagram illustrating all forces acting on the disk.



13. (20pts) A rod of length $L$ and mass $m$ is securely anchored at its center on a horizontal tabletop and can rotate freely about its central axis (as shown in the top-down view below). A force $F$ is applied at the right end of the rod to induce rotational motion.




a. (5pts) Using the variables $L$, $F$, and $\theta$, derive an expression for the magnitude of the net torque.

b. (5pts) Determine the direction of the net torque.

c. (10pts) Assuming the net torque remains constant for a time $t$, find the magnitude of the final angular velocity of the rod starting from rest. Use the variables $m$, $L$, $F$, $t$, and $\theta$ to derive an expression for your answer.

**Equation Sheet**

Gravitational acceleration: $g = 9.8$ m/s$^2$

Moment of inertia: $I = \sum m_i r_i^2$

The parallel-axis theorem: $I_{Total} = I_{CM} + Md^2$

Torque: $\vec{\tau} = \vec{r} \times \vec{F}$

Torque magnitude: $|\vec{\tau}| = |\vec{r} \times \vec{F}| = |\vec{r}||\vec{F}| \sin\theta$

Angular quantities: $s = R\theta$, $v = R\omega$, $a = R\alpha$

Translational/Linear dynamics: $\sum \vec{F} = m\vec{a}$

Rotational dynamics: $\sum \vec{\tau} = I\vec{\alpha}$

*The following four equations could also be applied to other directions, with a proper subscript modification:*

1D constant linear acceleration motion: $v_x = v_{0x} + a_x t$ and $x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$

Constant angular acceleration motion: $\omega_z = \omega_{0z} + \alpha_z t$ and $\theta_z = \theta_{0z} + \omega_{0z}t + \frac{1}{2}\alpha_z t^2$

Translational/Linear kinetic energy: $KE_T = \frac{1}{2}mv^2$

Rotational kinetic energy: $KE_R = \frac{1}{2}I\omega^2$

Angular momentum of a particle: $\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$

Angular momentum of a rigid body: $\vec{L} = I\vec{\omega}$

Torque and angular momentum: $\sum \vec{\tau} = \frac{d\vec{L}}{dt}$

Position vector of center of mass: $\vec{r}_{cm} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$

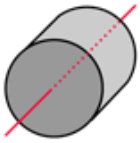
Velocity of center of mass: $\vec{V}_{cm} = \frac{\sum_i m_i \vec{V}_i}{\sum_i m_i}$

Relative motion: $\vec{V}_{A,B} = \vec{V}_{A,C} + \vec{V}_{C,B}$ and $\vec{V}_{A,B} = -\vec{V}_{B,A}$

Momentum: $\vec{p} = m\vec{v}$

Momentum conservation: $\sum \vec{p}_i = \sum \vec{p}_f$

Solid cylinder or disc, symmetry axis



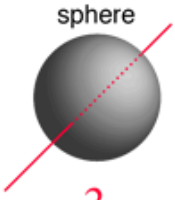
$$I = \frac{1}{2}MR^2$$

Hoop about symmetry axis



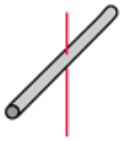
$$I = MR^2$$

Solid sphere



$$I = \frac{2}{5}MR^2$$

Rod about center



$$I = \frac{1}{12}ML^2$$